

```
!pip install -q pyspark
```

```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .master("local[*]") \
    .appName("ColabSpark") \
    .getOrCreate()

print("Spark Version:", spark.version)
```

Spark Version: 4.0.2

```
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True)

```
players = spark.read.csv("/content/drive/MyDrive/Datasets/WorldCupPlayers.csv", header = True)
```

```
players.show()
```

RoundID	MatchID	TeamInitials	CoachName	Line-up	ShirtNumber	PlayerName	Position	Event
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Alex THEPOT	GK	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Oscar BONFIGLIO	GK	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Marcel LANGILLER	NULL	G40'
201	1096	MEX	LUQUE Juan (MEX)	S	0	Juan CARRENO	NULL	G70'
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Ernest LIBERATI	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Rafael GARZA	C	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Andre MASCHINOT	NULL	G43' G87'
201	1096	MEX	LUQUE Juan (MEX)	S	0	Hilario LOPEZ	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Etienne MATTIER	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Dionisio MEJIA	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Marcel PINEL	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Felipe ROSAS	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Alex VILLAPLANE	C	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Manuel ROSAS	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Lucien LAURENT	NULL	G19'
201	1096	MEX	LUQUE Juan (MEX)	S	0	Jose RUIZ	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Marcel CAPELLE	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Alfredo SANCHEZ	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Augustin CHANTREL	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Efrain AMEZCUA	NULL	NULL

only showing top 20 rows

```
players.printSchema()
```

```
root
|-- RoundID: string (nullable = true)
|-- MatchID: string (nullable = true)
|-- TeamInitials: string (nullable = true)
|-- CoachName: string (nullable = true)
|-- Line-up: string (nullable = true)
|-- ShirtNumber: string (nullable = true)
|-- PlayerName: string (nullable = true)
|-- Position: string (nullable = true)
|-- Event: string (nullable = true)
```

```
num_rows = players.count()
num_columns = len(players.columns)

print(f"Number of rows: {num_rows}")
print(f"Number of columns: {num_columns}")
```

Number of rows: 37784
Number of columns: 9

```
k = players["PlayerName", "CoachName"]
k.show()
```

PlayerName	CoachName
Alex THEPOT	CAUDRON Raoul (FRA)
Oscar BONFIGLIO	LUQUE Juan (MEX)
Marcel LANGILLER	CAUDRON Raoul (FRA)
Juan CARRENO	LUQUE Juan (MEX)
Ernest LIBERATI	CAUDRON Raoul (FRA)
Rafael GARZA	LUQUE Juan (MEX)
Andre MASCHINOT	CAUDRON Raoul (FRA)
Hilario LOPEZ	LUQUE Juan (MEX)
Etienne MATTLER	CAUDRON Raoul (FRA)
Dionisio MEJIA	LUQUE Juan (MEX)
Marcel PINEL	CAUDRON Raoul (FRA)
Felipe ROSAS	LUQUE Juan (MEX)
Alex VILLAPLANE	CAUDRON Raoul (FRA)
Manuel ROSAS	LUQUE Juan (MEX)
Lucien LAURENT	CAUDRON Raoul (FRA)
Jose RUIZ	LUQUE Juan (MEX)
Marcel CAPELLE	CAUDRON Raoul (FRA)
Alfredo SANCHEZ	LUQUE Juan (MEX)
Augustin CHANTREL	CAUDRON Raoul (FRA)
Efrain AMEZCUA	LUQUE Juan (MEX)

only showing top 20 rows

```
filtered_players = players.filter(players['MatchID'] == 1096)
filtered_players.show()
```

RoundID	MatchID	TeamInitials	CoachName	Line-up	ShirtNumber	PlayerName	Position	Event
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Alex THEPOT	GK	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Oscar BONFIGLIO	GK	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Marcel LANGILLER	NULL	G40'
201	1096	MEX	LUQUE Juan (MEX)	S	0	Juan CARRENO	NULL	G70'
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Ernest LIBERATI	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Rafael GARZA	C	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Andre MASCHINOT	NULL	G43' G87'
201	1096	MEX	LUQUE Juan (MEX)	S	0	Hilario LOPEZ	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Etienne MATTLER	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Dionisio MEJIA	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Marcel PINEL	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Felipe ROSAS	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Alex VILLAPLANE	C	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Manuel ROSAS	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Lucien LAURENT	NULL	G19'
201	1096	MEX	LUQUE Juan (MEX)	S	0	Jose RUIZ	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Marcel CAPELLE	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Alfredo SANCHEZ	NULL	NULL
201	1096	FRA	CAUDRON Raoul (FRA)	S	0	Augustin CHANTREL	NULL	NULL
201	1096	MEX	LUQUE Juan (MEX)	S	0	Efrain AMEZCUA	NULL	NULL

only showing top 20 rows

```
num_filtered_rows = filtered_players.count()
print(f"Number of records with MatchID = 1096: {num_filtered_rows}")
```

Number of records with MatchID = 1096: 33

```
from pyspark.sql.functions import col

players_with_int_matchid = players.withColumn("MatchID_int", col("MatchID").cast("int"))

ordered_players_numeric = players_with_int_matchid.orderBy('MatchID_int')

ordered_players_numeric.show()
```

RoundID	MatchID	TeamInitials	CoachName	Line-up	ShirtNumber	PlayerName	Position	Event	MatchID_int
323	25	BRA	LAZARONI Sebastia...	S	1	TAFFAREL	GK	NULL	25
323	25	BRA	LAZARONI Sebastia...	S	21	MAURO GALVAO	NULL	Y50' 083'	25
323	25	ARG	BILARDO Carlos (ARG)	S	12	Sergio GOYCOCHEA	GK	Y87'	25
323	25	BRA	LAZARONI Sebastia...	S	2	JORGINHO	NULL	NULL	25
323	25	ARG	BILARDO Carlos (ARG)	S	4	Jose BASUALDO	NULL	NULL	25
323	25	BRA	LAZARONI Sebastia...	S	3	RICARDO GOMES	C	R85'	25

323	25	ARG	BILARDO Carlos (ARG)	S	7	Jorge BURRUCHAGA	NULL	NULL	25
323	25	BRA	LAZARONI Sebastia...	S	4	DUNGA	NULL	NULL	25
323	25	ARG	BILARDO Carlos (ARG)	S	8	Claudio CANIGGIA	NULL	G81'	25
323	25	BRA	LAZARONI Sebastia...	S	5	ALEMAO	NULL	O83'	25
323	25	ARG	BILARDO Carlos (ARG)	S	10	Diego MARADONA	C	NULL	25
323	25	BRA	LAZARONI Sebastia...	S	6	BRANCO	NULL	NULL	25
323	25	ARG	BILARDO Carlos (ARG)	S	14	Ricardo GIUSTI	NULL	Y28'	25
323	25	BRA	LAZARONI Sebastia...	S	8	VALDO	NULL	NULL	25
323	25	ARG	BILARDO Carlos (ARG)	S	15	Pedro MONZON	NULL	Y27'	25
323	25	BRA	LAZARONI Sebastia...	S	9	CARECA	NULL	NULL	25
323	25	ARG	BILARDO Carlos (ARG)	S	16	Julio OLARTICOECHEA	NULL	NULL	25
323	25	BRA	LAZARONI Sebastia...	S	15	MULLER	NULL	NULL	25
323	25	ARG	BILARDO Carlos (ARG)	S	19	Oscar RUGGERI	NULL	NULL	25
323	25	BRA	LAZARONI Sebastia...	S	19	RICARDO ROCHA	NULL	Y40'	25

only showing top 20 rows

```
filtered_position_event = players.filter((players['Position'] == 'C') & (players['Event'] == "G40'"))
filtered_position_event.show()
```

RoundID	MatchID	TeamInitials	CoachName	Line-up	ShirtNumber	PlayerName	Position	Event
201	1089	PAR	DURAND LAGUNA Jos...	S	0	Luis VARGAS PENA	C	G40'
429	1175	HUN	DIETZ Karoly (HUN)	S	0	Gyorgy SAROSI	C	G40'

```
unique_match_records = players.dropDuplicates(['MatchID'])
unique_match_records.show()
```

RoundID	MatchID	TeamInitials	CoachName	Line-up	ShirtNumber	PlayerName	Position	Event
293	1012	TCH	VENGLOS Jozef (SVK)	S	21	Zdenek HRUSKA	GK	NULL
323	102	CMR	NEPOMNYASHCHI Val...	S	16	Thomas NKONO	GK	NULL
751	103	ENG	ROBSON Bobby (ENG)	S	1	Peter SHILTON	GK	NULL
293	1044	YUG	MILJANIC Miljan (...)	S	1	Dragan PANTELIC	GK	NULL
293	1051	SCO	STEIN Jock (SCO)	S	1	Alan ROUGH	GK	NULL
293	1054	URS	BESKOV Konstantin...	S	1	Rinat DASAEV	GK	NULL
293	1055	POL	PIECHNICZEK Anton...	S	1	Jozef MLYNARCZYK	GK	NULL
294	1058	POL	PIECHNICZEK Anton...	S	1	Jozef MLYNARCZYK	GK	NULL
293	1071	URS	BESKOV Konstantin...	S	1	Rinat DASAEV	GK	NULL
322	108	CMR	NEPOMNYASHCHI Val...	S	16	Thomas NKONO	GK	Y68'
201	1084	ARG	OLAZAR Francisco ...	S	0	Angel BOSSIO	GK	NULL
201	1085	ARG	OLAZAR Francisco ...	S	0	Angel BOSSIO	GK	NULL
201	1086	ARG	OLAZAR Francisco ...	S	0	Angel BOSSIO	GK	NULL
405	1087	URU	SUPPICI Alberto (...)	S	0	Enrique BALLESTRERO	GK	NULL
202	1088	ARG	OLAZAR Francisco ...	S	0	Juan BOTASSO	GK	NULL
201	1089	PAR	DURAND LAGUNA Jos...	S	0	Pedro BENITEZ	GK	NULL
201	1090	USA	MILLAR Bob (USA)	S	0	Jimmy DOUGLAS	GK	NULL
201	1091	BRA	DE CARVALHO Pinda...	S	0	VELLOSO	GK	NULL
201	1092	YUG	SIMONOVIC Bosko (...)	S	0	Milovan JAKSIC	GK	NULL
201	1093	YUG	SIMONOVIC Bosko (...)	S	0	Milovan JAKSIC	GK	NULL

only showing top 20 rows

```
players.createOrReplaceTempView("players_table")
```

```
sql_query = "SELECT PlayerName, CoachName FROM players_table"
player_coach_df = spark.sql(sql_query)
player_coach_df.show()
```

PlayerName	CoachName
Alex THEPOT	CAUDRON Raoul (FRA)
Oscar BONFIGLIO	LUQUE Juan (MEX)
Marcel LANGILLER	CAUDRON Raoul (FRA)
Juan CARRENO	LUQUE Juan (MEX)
Ernest LIBERATI	CAUDRON Raoul (FRA)
Rafael GARZA	LUQUE Juan (MEX)
Andre MASCHINOT	CAUDRON Raoul (FRA)
Hilario LOPEZ	LUQUE Juan (MEX)
Etienne MATTLER	CAUDRON Raoul (FRA)
Dionisio MEJIA	LUQUE Juan (MEX)
Marcel PINEL	CAUDRON Raoul (FRA)
Felipe ROSAS	LUQUE Juan (MEX)
Alex VILLAPLANE	CAUDRON Raoul (FRA)

```

| Manuel ROSAS | LUQUE Juan (MEX) |
| Lucien LAURENT | CAUDRON Raoul (FRA) |
| Jose RUIZ | LUQUE Juan (MEX) |
| Marcel CAPELLE | CAUDRON Raoul (FRA) |
| Alfredo SANCHEZ | LUQUE Juan (MEX) |
| Augustin CHANTREL | CAUDRON Raoul (FRA) |
| Efrain AMEZCUA | LUQUE Juan (MEX) |
+-----+
only showing top 20 rows

```

```
from pyspark.sql.functions import lit
```

```

players_with_year = players.withColumn("year", lit("1930"))
players_with_year.show(5)

```

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|RoundID|MatchID|TeamInitials|CoachName|Line-up|ShirtNumber|PlayerName|Position|Event|year|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Alex THEPOT | GK | NULL | 1930 |
| 201 | 1096 | MEX | LUQUE Juan (MEX) | S | 0 | Oscar BONFIGLIO | GK | NULL | 1930 |
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Marcel LANGILLER | NULL | G40' | 1930 |
| 201 | 1096 | MEX | LUQUE Juan (MEX) | S | 0 | Juan CARRENO | NULL | G70' | 1930 |
| 201 | 1096 | FRA | CAUDRON Raoul (FRA) | S | 0 | Ernest LIBERATI | NULL | NULL | 1930 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
only showing top 5 rows

```

```
mtcars = spark.read.csv("/content/drive/MyDrive/Datasets/mtcars.csv", header = True)
```

```
mtcars.show()
```

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|model|mpg|cyl|disp|hp|drat|wt|qsec|vs|am|gear|carb|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|Mazda RX4|21|6|160|110|3.9|2.62|16.46|0|1|4|4|
|Mazda RX4 Wag|21|6|160|110|3.9|2.875|17.02|0|1|4|4|
|Datsun 710|22.8|4|108|93|3.85|2.32|18.61|1|1|4|1|
|Hornet 4 Drive|21.4|6|258|110|3.08|3.215|19.44|1|0|3|1|
|Hornet Sportabout|18.7|8|360|175|3.15|3.44|17.02|0|0|3|2|
|Valiant|18.1|6|225|105|2.76|3.46|20.22|1|0|3|1|
|Duster 360|14.3|8|360|245|3.21|3.57|15.84|0|0|3|4|
|Merc 240D|24.4|4|146.7|62|3.69|3.19|20|1|0|4|2|
|Merc 230|22.8|4|140.8|95|3.92|3.15|22.9|1|0|4|2|
|Merc 280|19.2|6|167.6|123|3.92|3.44|18.3|1|0|4|4|
|Merc 280C|17.8|6|167.6|123|3.92|3.44|18.9|1|0|4|4|
|Merc 450SE|16.4|8|275.8|180|3.07|4.07|17.4|0|0|3|3|
|Merc 450SL|17.3|8|275.8|180|3.07|3.73|17.6|0|0|3|3|
|Merc 450SLC|15.2|8|275.8|180|3.07|3.78|18|0|0|3|3|
|Cadillac Fleetwood|10.4|8|472|205|2.93|5.25|17.98|0|0|3|4|
|Lincoln Continental|10.4|8|460|215|3|5.424|17.82|0|0|3|4|
|Chrysler Imperial|14.7|8|440|230|3.23|5.345|17.42|0|0|3|4|
|Fiat 128|32.4|4|78.7|66|4.08|2.2|19.47|1|1|4|1|
|Honda Civic|30.4|4|75.7|52|4.93|1.615|18.52|1|1|4|2|
|Toyota Corolla|33.9|4|71.1|65|4.22|1.835|19.9|1|1|4|1|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
only showing top 20 rows

```

```
from pyspark.sql.functions import col
```

```
mtcars_casted_mpg = mtcars.withColumn("mpg_float", col("mpg").cast("float"))
```

```

top_five_mpg = mtcars_casted_mpg.orderBy(col("mpg_float").desc()).limit(5)
top_five_mpg.show()

```

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|model|mpg|cyl|disp|hp|drat|wt|qsec|vs|am|gear|carb|mpg_float|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|Toyota Corolla|33.9|4|71.1|65|4.22|1.835|19.9|1|1|4|1|33.9|
|Fiat 128|32.4|4|78.7|66|4.08|2.2|19.47|1|1|4|1|32.4|
|Honda Civic|30.4|4|75.7|52|4.93|1.615|18.52|1|1|4|2|30.4|
|Lotus Europa|30.4|4|95.1|113|3.77|1.513|16.9|1|1|5|2|30.4|
|Fiat X1-9|27.3|4|79|66|4.08|1.935|18.9|1|1|4|1|27.3|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

```

mpg_less_than_18 = mtcars_casted_mpg.filter(mtcars_casted_mpg['mpg_float'] < 18)
mpg_less_than_18.show()

```

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|      model| mpg|cyl| disp| hp|drat|   wt|  qsec|  vs| am|gear|carb|mpg_float|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|      Duster 360|14.3|  8|  360|245|3.21|  3.57|15.84|  0|  0|  3|  4|    14.3|
|      Merc 280C|17.8|  6|167.6|123|3.92|  3.44| 18.9|  1|  0|  4|  4|    17.8|
|      Merc 450SE|16.4|  8|275.8|180|3.07|  4.07| 17.4|  0|  0|  3|  3|    16.4|
|      Merc 450SL|17.3|  8|275.8|180|3.07|  3.73| 17.6|  0|  0|  3|  3|    17.3|
|      Merc 450SLC|15.2|  8|275.8|180|3.07|  3.78|  18|  0|  0|  3|  3|    15.2|
|  Cadillac Fleetwood|10.4|  8|  472|205|2.93|  5.25|17.98|  0|  0|  3|  4|    10.4|
| Lincoln Continental|10.4|  8|  460|215|   3|5.424|17.82|  0|  0|  3|  4|    10.4|
| Chrysler Imperial|14.7|  8|  440|230|3.23|5.345|17.42|  0|  0|  3|  4|    14.7|
| Dodge Challenger|15.5|  8|  318|150|2.76|  3.52|16.87|  0|  0|  3|  2|    15.5|
| AMC Javelin|15.2|  8|  304|150|3.15|3.435|17.3|  0|  0|  3|  2|    15.2|
| Camaro Z28|13.3|  8|  350|245|3.73|  3.84|15.41|  0|  0|  3|  4|    13.3|
| Ford Pantera L|15.8|  8|  351|264|4.22|  3.17| 14.5|  0|  1|  5|  4|    15.8|
| Maserati Bora| 15|  8|  301|335|3.54|  3.57| 14.6|  0|  1|  5|  8|    15.0|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

```
from pyspark.sql.functions import col
```

```
mtcars_with_wtTon = mtcars.withColumn("wtTon", col("wt").cast("float") * 0.45)
mtcars_with_wtTon.show(5)
```

```

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|      model| mpg|cyl| disp| hp|drat|   wt|  qsec|  vs| am|gear|carb|          wtTon|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|      Mazda RX4|  21|  6| 160|110| 3.9|  2.62|16.46|  0|  1|  4|  4| 1.178999948501587|
| Mazda RX4 Wag|  21|  6| 160|110| 3.9|  2.875|17.02|  0|  1|  4|  4|    1.29375|
| Datsun 710|22.8|  4| 108| 93|3.85|  2.32|18.61|  1|  1|  4|  1| 1.043999969959259|
| Hornet 4 Drive|21.4|  6| 258|110|3.08|3.215|19.44|  1|  0|  3|  1|1.4467499613761903|
|Hornet Sportabout|18.7|  8| 360|175|3.15|  3.44|17.02|  0|  0|  3|  2|1.5480000257492066|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

only showing top 5 rows

```
from pyspark.sql.functions import col, avg
```

```
average_wt_by_cyl = mtcars.groupBy('cyl') \
                          .agg(avg(col('wt').cast('float')).alias('average_weight')) \
                          .orderBy('cyl')
```

```
average_wt_by_cyl.show()
```

```

+---+-----+
|cyl| average_weight|
+---+-----+
|  4|2.2857272841713647|
|  6| 3.117142847606114|
|  8|3.9992142575127736|
+---+-----+

```

```
mtcars.createOrReplaceTempView("mtcars_table")
```

```
sql_query_cyl_filter = "SELECT gear FROM mtcars_table WHERE cyl BETWEEN 4 AND 9"
gears_filtered_by_cyl = spark.sql(sql_query_cyl_filter)
gears_filtered_by_cyl.show()
```

```

+----+
|gear|
+----+
|  4|
|  4|
|  4|
|  3|
|  3|
|  3|
|  3|
|  4|
|  4|
|  4|
|  4|
|  3|
|  3|
|  3|
|  3|
|  3|
|  3|

```

```
| 4 |  
| 4 |  
| 4 |  
+----+
```

only showing top 20 rows